

Circular choosability of planar graphs

OPG-37619

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $G = (V, E)$ be a graph. If p and q are two integers, a (p, q) -colouring of G is a function c from V to $\{0, \dots, p-1\}$ such that $q \leq |c(u) - c(v)| \leq p - q$ for each edge $uv \in E$. Given a list assignment L of G , i.e. a mapping that assigns to every vertex v a set of non-negative integers, an L -colouring of G is a mapping $c : V \rightarrow \mathbb{N}$ such that $c(v) \in L(v)$ for every $v \in V$. A list assignment L is a t - (p, q) -list-assignment if $L(v) \subseteq \{0, \dots, p-1\}$ and $|L(v)| \geq tq$ for each vertex $v \in V$. Given such a list assignment L , the graph G is (p, q) - L -colourable if there exists a (p, q) - L -colouring c , i.e. c is both a (p, q) -colouring and an L -colouring. For any real number $t \geq 1$, the graph G is t - (p, q) -choosable if it is (p, q) - L -colourable for every t - (p, q) -list-assignment L . Last, G is circularly t -choosable if it is t - (p, q) -choosable for any p, q . The circular choosability (or circular list chromatic number or circular choice number) of G is

$$cch(G) := \inf\{t \geq 1 : G \text{ is circularly } t\text{-choosable}\}.$$

Problem What is the best upper bound on circular choosability for planar graphs?

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